

```
SEMESTER 3 > C 2b.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  int main(){
4      char name[10];
5      strcpy(name,"ABCDEFGHJKLMNOP");
6      printf("Value: %s\n", name);
7      return 0;
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Value: ABCDEFGHJKLMNOP
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2d.c > main()
1  #include <stdio.h>
2  int main(){
3      char username[10];
4      printf("Enter username: ");
5      scanf("%9s", username); // Restricted to 9 chars + '\0'
6      printf("Username: %s\n", username);
7      return 0;
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Enter username: Amrit
Username: Amrit
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2f.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  struct StudentData {
4      char buffer [8];
5      char status [8];
6  };
7  int main(){
8      struct StudentData data;
9      strcpy(data.status, "SAFE");
10     printf("Address of buffer: %p\n", (void *)data.buffer);
11     printf("Address of status: %p\n", (void *)data.status);
12     strcpy(data.buffer, "AAAAAAAHACKED");
13     printf("After overflow, Status = %s\n", data.status);
14     return 0;
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Address of buffer: 0061FF10
Address of status: 0061FF18
After overflow, Status = HACKED
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2e.c > main()
1  #include <stdio.h>
2  int main(){
3      char username[10];
4      printf("Enter username: ");
5      fgets(username,sizeof(username),stdin);
6      printf("Username: %s", username);
7      return 0;
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Enter username: Amrit Mahajan
Username: Amrit Mah
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2a.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  int main(){
4      char name[10];
5      strcpy(name, "Amrit");
6      printf("%s | Buffer size: %zu bytes\n", name, sizeof(name));
7      return 0;
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Amrit | Buffer size: 10 bytes
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2c.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  struct StudentData {
4      char buffer [8];
5      char status [8];
6  };
7  int main(){
8      struct StudentData data;
9      strcpy(data.status, "SAFE");
10     printf("BEFORE: Status = %s\n", data.status);
11     strcpy(data.buffer, "AAAAAAAHACKED"); // oversized copy into buffer
12     printf("AFTER: Buffer = %s\n", data.buffer);
13     printf("AFTER: Status = %s\n", data.status);
14     return 0;
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
BEFORE: Status = SAFE
AFTER: Buffer = AAAAAAAHACKED
AFTER: Status = HACKED
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```

```
SEMESTER 3 > C 2g.c > main()
1  #include <stdio.h>
2  #include <string.h>
3  int main(){
4      char buffer[10];
5      snprintf(buffer,sizeof(buffer), "%s", "ABCDEFGHJKLMNOPQRSTUVWXYZ");
6      printf("Stored value: %s\n", buffer);
7      return 0;
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\"
Stored value: ABCDEFGHI
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3> |
```